

| Phase | Task                                                                                                                                                       | Week        |             |             |   |        |             |   |   |       |     |             |    |    |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|-------------|---|--------|-------------|---|---|-------|-----|-------------|----|----|
|       |                                                                                                                                                            | 1           | 2           | 3           | 4 | 5      | 6           | 7 | 8 | 9     | 10  | 11          | 12 | 13 |
| 1     | 1. Project plan                                                                                                                                            |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Complete and submit a Project Plan</i>                                                                                                                  | ALL         |             |             |   |        |             |   |   |       |     |             |    |    |
|       | 2. Set up Raspberry Pi as Server and confirmed communication                                                                                               |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Set up Raspberry Pi as a server</i>                                                                                                                     | Fergus      |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Confirm communication between client computer and server (HTTP/SSH)</i>                                                                                 | Fergus      |             |             |   |        |             |   |   |       |     |             |    |    |
|       | 3. STM32 Programming via HAL                                                                                                                               |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Explain common CAN messaging protocol (see document)</i>                                                                                                | Nigel       |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Implement the CAN messaging protocol by properly setting the filters (code demo) on STM32</i>                                                           |             | Nigel       |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Implement Wiring on STM32, push buttons/LED indicators (possible COVID restrictions)</i>                                                                |             | Nigel       |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Able to read CAN messages placed on the CAN bus (Raspberry Pi) and detect up/down request, floor request, door open/close, location of elevator car</i> |             | Ryan        |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Develop and implement a test plan to confirm functionality of software/hardware</i>                                                                     | ALL         |             |             |   |        |             |   |   |       |     |             |    |    |
|       | 3. Controls / indicators (virtual and buttons)                                                                                                             |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Controls (up/down) built in HTML (possibly add CSS if students work ahead)</i>                                                                          | Fergus/Ryan |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Elevator Car Controls (up/down/floor number) built in HTML (possibly CSS if students work ahead)</i>                                                    | Fergus/Ryan |             |             |   |        |             |   |   |       |     |             |    |    |
| 2     | 4. Finite State Machine (recommended)                                                                                                                      |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Develop Finite State Machine Logic for Elevator Operation</i>                                                                                           | Ryan        |             |             |   |        |             |   |   |       |     |             |    |    |
|       | 1. User Interface / Client Program                                                                                                                         |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Fully working and completed front end Graphical User Interface /Webpage to control elevator</i>                                                         |             |             |             |   |        | Fergus      |   |   |       |     |             |    |    |
|       | <i>User action buttons/icons implemented</i>                                                                                                               |             | Fergus      |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Indicators/lights etc. implemented</i>                                                                                                                  |             |             | Fergus/Ryan |   |        |             |   |   |       |     |             |    |    |
|       | <i>User action buttons developed to make changes to UI that are reflected in the database</i>                                                              |             | Fergus/Ryan |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>User indicator buttons change - changes in database are reflected in UI</i>                                                                             |             |             |             |   |        | Fergus/Ryan |   |   |       |     |             |    |    |
|       | 2. Database Repository                                                                                                                                     |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Database set up and accepting values/commands from website</i>                                                                                          |             |             |             |   |        | Ryan        |   |   |       |     |             |    |    |
|       | <i>Database has fields for logging information placed on the CAN bus</i>                                                                                   |             |             |             |   |        | Ryan        |   |   |       |     |             |    |    |
|       | 3. Diagnostic Reports                                                                                                                                      |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Produced based on content in database</i>                                                                                                               |             |             | Ryan/Nigel  |   |        |             |   |   |       |     |             |    |    |
|       | <i>Displayed on webpage</i>                                                                                                                                |             |             | Nigel       |   |        |             |   |   |       |     |             |    |    |
| 3     | 4. Server Program                                                                                                                                          |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Users are authorized/authenticated based on username/password</i>                                                                                       |             |             |             |   | Fergus |             |   |   |       |     |             |    |    |
|       | <i>Server program provides logging data/accepts commands only from authorized users</i>                                                                    |             |             |             |   | Fergus |             |   |   |       |     |             |    |    |
|       | 5. Announce floor numbers (Function of the Client or Server Program) for users with Visual Impairments                                                     |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Implement a Server Program for announcing the floor numbers</i>                                                                                         |             |             |             |   |        | Nigel       |   |   |       |     |             |    |    |
|       | <i>Speaker on elevator used to announce the current/next floor (speaker already connected to system)</i>                                                   |             |             |             |   |        | Nigel       |   |   |       |     |             |    |    |
|       | 6. Sabbath mode (Function of the Client Program)                                                                                                           |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Users may operate the elevator without having to push buttons (e.g. verbally, based on schedule/time)</i>                                               |             |             |             |   |        | Ryan        |   |   |       |     |             |    |    |
|       | 1. CAN/LAN network communication                                                                                                                           |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Research and Implement a CAN/LAN communication bridge</i>                                                                                               |             |             |             |   |        |             |   |   |       |     | Nigel/Ryan  |    |    |
|       | <i>Ensure activity in CAN network changes values in database</i>                                                                                           |             |             |             |   |        |             |   |   |       |     | Nigel/Ryan  |    |    |
|       | <i>Changes in User Interface/Website changes value in database and generate CAN messages</i>                                                               |             |             |             |   |        |             |   |   |       |     | Fergus/Ryan |    |    |
|       | <i>CAN Messages result in changes to database and GUI/Indicators</i>                                                                                       |             |             |             |   |        |             |   |   |       |     | ALL         |    |    |
|       | <i>Commands from CAN/LAN networks are sent to the controller and move the elevator / indicate floor number</i>                                             |             |             |             |   |        |             |   |   |       |     | ALL         |    |    |
|       | 2. Controller Software Written                                                                                                                             |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>CAN Supervisor (Raspberry Pi) relays messages between CAN/LAN Network (see sample program on Rpi)</i>                                                   |             |             |             |   |        |             |   |   |       |     | Nigel/Ryan  |    |    |
|       | 3. Finite State Machine logic written                                                                                                                      |             |             |             |   |        |             |   |   |       |     |             |    |    |
|       | <i>Develop and Implement a finte state machine (using C/C+, PHP, Python to implement logic)</i>                                                            |             |             | Nigel       |   |        |             |   |   |       |     |             |    |    |
|       | 4. Additional capability that shows ingenuity                                                                                                              |             |             |             |   |        |             |   |   | Nigel |     |             |    |    |
|       | 5. Ideas for Capstone (at least 3 per student)                                                                                                             |             |             | ALL         |   |        |             |   |   |       |     |             |    |    |
|       | 6. Final Presentation & Demonstration                                                                                                                      |             |             |             |   |        |             |   |   |       | ALL |             |    |    |

\*Project ends on Monday of week 14